HW 5.1

5.1.2 a) Total Characters = Digits + Letters + Special Characters

$= 10 + 26 + 4$

$= 40$

Number of Passwords = 40 down till there are 6 characters

$= 40(39)(38)(37)(36)(35)$

$= 2,763,633,600$

b) total Char = 40

$P_7 = (40)(39)(38)(37)(36)(35)(34) = 93,963,542,400$

$P_8 = (40)(39)(38)(37)(36)(35)(34)(33) = 3,100,796,899,200$

$P_9 = (40)(39)(38)(37)(36)(35)(34)(33)(32) = 99,225,500,774,400$

$P_7 + P_8 + P_9 = 102,420,261,216,000$

c) first character = total(40) - Letters(26)

$= 14$

$P_7 = 14(39)(38)(37)(36)(35)(34) = 32,887,239,840$

$P_8 = 14(39)(38)(37)(36)(35)(34)(33) = 1,085,278,914,720$

$P_9 = 14(39)(38)(37)(36)(35)(34)(33)(32) = 34,728,925,271,040$

$P_7 + P_8 + P_9 = 35,847,091,425,600$

5.2 HW

5.2.2 a) Show a bijection between P_6 and B^3

P_6 = palindromes of length 6.

B^3: bit strings of length 3.

By definition: length: $P_6 \to B^3$

let x,y $\exists x \exists y \in P_6$ Length(abccba) = abc

Ex. Length(101101) = 101 where $a,b,c \in \{0,1\}$

Length(001100) = 001

Let Length($a_1 b_1 c_1 c_1 b_1 a_1$) = Length($a_2 b_2 c_2 c_2 b_2 a_2$)

$a_1 b_1 c_1 = a_2 b_2 c_2$

$a_1 = a_2$ $b_1 = b_2$ $c_1 = c_2$

length is one to one and length is a bijection.

b) What is $|P_6|$?

$|P_6| = |B^3|$

B has 3 places, with each place having 2 options $\{0,1\}$

$|B^3| = 2^3 = 8$ $|P_6| = 8$

c) Define $F: P_7 \to B^4$ as

F(abcddcba) = abcd

Ex. F(00011000) = 0001

F is a bijection so $|P_7| = |B^4|$

$B = \{0,1\}$ - 2 options w/ 4 places

$|B^4| = 2^4 = 16$

$|P_7| = 16$

5.3 HW

5.3.2 set $\{a, b, c\}$ with length 10. No consecutive characters.

First character has 3 choices $\{a, b, c\}$
2nd - 10 has 2 choices. Can not be previous character.

Number of strings $= 3 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2$
$= 3 * 2^9$
$= 1536$

5.4 HW

5.4.3 a) 10 people - no constraints

$10! = 3628800$

b) P and VP act as 1 member with 2 interchangeable options. In this way there are 9 members with twice as many permutations

$= 2 * 9!$

$= 725760$

c) President and secretary act as 1 member with 2 options. VP and treasurer act as 1 member with 2 options.

$= 2 * 2 * 8!$

$= 161{,}280$

5.5 HW

5.5.1 a) What is the value of f on input $(12,1,3,15,9)$?

Number of arrangements: $5!$

Number of permutations: $5!$

$$\frac{5!}{5} =$$

$$= 4!$$

Number of permutations $\{12,1,3,15,9\} = 24$

b) is $(12,3,12,4,19)$ a 5-permutation?

It is not a 5-permutation because the element 12 is repeated.

c) How many permutations are mapped onto the subset $\{12,3,13,4,19\}$

$$\text{number of permutation} = \frac{5!}{5(5-5)!} = \frac{5!}{5} = 24$$

5.6 HW

5.6.2 a) 120 = Pianists

30/120 past 1st round

$P(120, 30)$ or $P\binom{120}{30}$

$$= \frac{120!}{30!(120-30)!} = \frac{120!}{30!(90!)} = 1.6974539e^{28}$$

b) 30 remaining pianists

5 total placings that are distinct

$$P\binom{30}{5} = 30 * 29 * 28 * 27 * 26$$

$= 17100720$ ways

5.7 HW

5.7.2 a) total # of cards in deck : 52
total # of distinct clubs : 13

$$52 \text{ cards} = C(n,r) = \frac{n!}{(r!(n-r)!)}$$

$n = 52$
$r = 5$

$$= \frac{52!}{5!(52-5)!}$$

$$= \frac{52!}{5!(47!)} = 2598960 \text{ total number of 5 card sets}$$

without a club: $52 - 13 = 39$

$$C_2(n,r) = \frac{39!}{5!(39-5)!} = 575757$$

set of 5 cards with a club $= 2598960 - 575757$
$= 2023203$

b) Number of Ranks = 13

5 card hands with 13 rank combinations.

$$C(n,r) = \frac{13!}{5!(13-5)!} = \frac{13!}{5!(8)!} = 1287$$

possible ways to sort 4 suits with 5 cards $= 4^5$

$= 1287 * 4^5$
$= 1287 * 1024$
$= 1317888$

$\therefore$ Number of 5 card sets with at least 2 cards of the same rank $= 2598960 - 1317888$
$= 1,281,072$

5.9 HW

5.9.2 a) total number of ways $\begin{bmatrix} 15(6-1) \\ 15 \end{bmatrix} \rightarrow {}^{20}C_{15} = 15504$ ways

b) 3 must be chocolate

Total (15) - 3

$\begin{bmatrix} 12+(6-1) \\ 6-1 \end{bmatrix} = {}^{17}C_{5}$

$\sum_{k=3}^{15} \binom{19-k}{4} = 15504 - \sum_{k=0}^{2} \binom{19-k}{4}$ = 6188 ways

c) if only 2 can be sugar.

total # of ways remaining = 15504 - 6188 ways = 9316

d)

5.10 HW

5.10.5 a) total different ways $= {}^{25}C_{10}$ $\quad mCn$

$$= \frac{25!}{10!(25-10)!}$$

$$= \frac{25!}{10!(15)!}$$

$$= 3268760$$

b) total ways $= P(m,n)$

$$= \frac{25!}{(25-10)!}$$

$$= \frac{25!}{15!} = 11{,}861{,}676{,}288{,}000$$

5.12 HW

5.12.2 a) ${}^{n-1}C_{r-1}$ "No restrictions"

$r = 6$, $n = 25$

${}^{(25-1)}C_{(6-1)} = {}^{24}C_5$

$= \frac{24!}{5!(24-5)!}$

$= \frac{24!}{5!(19)!} = 42504$

b) $25 - 18 = 7$

${}^{(7-1)}C_{(6-1)} = {}^{6}C_5 = \frac{6!}{5!(6-5)!}$

$= \frac{6!}{5!(1!)} = 6$

c) $x \leq 3 = 25 - 3$

$= 22$

${}^{(22-1)}C_{(6-1)} = {}^{21}C_5 = \frac{21!}{5!(16)!} = 20349$

and

$x \leq 10 = 25 - 10$

$= 15$

$= {}^{(15-1)}C_{(6-1)} = {}^{14}C_5 = 2002$

${}^{21}C_5 - {}^{14}C_5 = 20349 - 2002$

$= 18347$

d) ${}^{6+(20-1)}C_{20} + {}^{6+(6-1)}C_6 - {}^{6+(12-1)}C_{12} - {}^{6+(14-1)}C_{14}$

$= {}^{25}C_{20} + {}^{11}C_6 - {}^{17}C_{12} - {}^{19}C_{14}$

$= \frac{25!}{20!(5!)} + \frac{11!}{6!(5!)} - \frac{17!}{12!(5!)} - \frac{19!}{14!(5!)}$

$= 53130 + 462 - 6188 - 11{,}628$

$= 35776$